MotoTrack Professional Features Guide

This document explains the next professional-level features you can add to your Laravel + Supabase project.

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1. API Routes

What are API Routes?

API routes allow other applications, mobile apps, or frontend frameworks to communicate with your Laravel backend.

Instead of returning Blade views/webpages, API routes return JSON data.

File location:

```
routes/api.php
```

Example API Route

```
use App\Models\Product;
use Illuminate\Support\Facades\Route;

Route::get('/products', function () {
    return Product::all();
});
```

Visiting:

```
/api/products
```

Returns:

```
[
  {
    "id": 1,
    "name": "Helmet",
    "price": 2500
```

```
  }
]
```

Recommended APIs for MotoTrack

Products

```
GET /api/products
POST /api/products
PUT /api/products/{id}
DELETE /api/products/{id}
```

Repairs

```
GET /api/repairs
POST /api/repairs
```

Orders

```
GET /api/orders
POST /api/orders
```

Customers

```
GET /api/customers
```

Benefits

- Mobile app support
- React/Vue frontend support
- Third-party integrations
- Better scalability
- More professional architecture

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2. Activity Logging

What is Activity Logging?

Activity logging records who performed actions in the system.

Examples:

```
Admin updated inventory
Customer created preorder
Owner changed repair status
```

Recommended Package

Install:

```
composer require spatie/laravel-activitylog
```

Documentation: https://spatie.be/docs/laravel-activitylog

Example Usage

```
activity()
    ->causedBy(auth()->user())
    ->log('Updated inventory');
```

Recommended Logs for MotoTrack

Inventory

- Product added
- Product updated
- Product deleted
- Stock changes

Repairs

- Repair created
- Status updated
- Repair completed

Orders

- Order placed
- Payment completed
- Order cancelled

Users

- Login
- Logout
- Role changes

Benefits

- Audit trail
- Employee tracking
- Easier debugging
- Better security
- Professional enterprise feature

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3. Dashboard Analytics

What are Dashboard Analytics?

Dashboard analytics display statistics and charts inside the admin dashboard.

Recommended Metrics

Sales

- Daily sales
- Monthly revenue
- Total income

Products

- Best-selling products
- Low stock alerts
- Inventory count

Repairs

- Pending repairs
- Completed repairs
- Repair trends

Customers

- Total customers
- New registrations
- Active users

Recommended Chart Libraries

Chart.js

https://www.chartjs.org

ApexCharts

https://apexcharts.com

Example Query

```
$totalSales = Order::sum('total');
```

Example Chart Data

```
$salesPerMonth = Order::selectRaw('MONTH(created_at) as month, SUM(total) as
total')
    ->groupBy('month')
    ->get();
```

Benefits

- Better business insights
- Professional admin dashboard
- Better client presentation
- Easier decision making

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4. Automated Testing

What are Tests?

Tests automatically check if features work correctly.

This prevents updates from accidentally breaking the application.

Laravel Testing

Laravel already includes:

- PHPUnit
- Pest

Example Test

```
it('creates an order', function () {
    $response = $this->post('/orders', [
        'product_id' => 1
    ]);

    $response->assertStatus(200);
});
```

Running Tests

```
php artisan test
```

Recommended Tests for MotoTrack

Authentication

- Login works
- Registration works
- Unauthorized access blocked

Orders

- Order creation
- Payment validation
- Checkout process

Repairs

- Repair submission
- Status updates

Inventory

- Product creation
- Stock updates

- Product deletion

Benefits

- Safer updates
- Fewer bugs
- Easier maintenance
- More professional codebase

Recommended Priority for MotoTrack

Highest Priority

1. Dashboard Analytics
2. Activity Logging

These immediately improve professionalism and portfolio quality.

Medium Priority

1. API Routes

Useful for future mobile apps or frontend frameworks.

Advanced Priority

1. Automated Testing

Very important for long-term maintenance and client projects.

Suggested Modern Architecture

```
Frontend:
- Blade / Livewire
or
- React / Vue

Backend:
- Laravel API

Database:
- Supabase PostgreSQL
```

```
Hosting:
- Laravel Cloud

Payments:
- Xendit

Analytics:
- Chart.js / ApexCharts

Authentication:
- Laravel Breeze / Sanctum
```

Final Notes

MotoTrack already has strong foundations:

- Role management
- Inventory system
- Repairs management
- Payments integration
- Notifications
- PostgreSQL support

The next step is adding professional SaaS-level features.

Focus on:

1. Better admin dashboard
2. Logging and audit trails
3. APIs
4. Automated testing

These features will make the project much stronger for:

- Freelancing
- Portfolio presentation
- Client projects
- SaaS development
- Long-term scalability